

Name:

Points: /20

Linux Server & Bash Programming — Final Exam Retake

2025-01-22

Instructions

1. No computers, no phones, no notes
2. No talking with other people in the room
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. For each question, you'll need to answer the question by crossing with an **X** the right answer.
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

Each question gives **1 point**. There are **20 questions**, for a total of maximum **20 points**.

Each correct answer gives **1 point**.

Grading

Points	Grade
0 - 9	IG
10 - 17	G
18 - 20	VG

1. Difficult
2. More fun
- ✓ 3. A file ✓
4. A terminal
5. A shell

2 How many addresses can I have in a /28 network? (NB: addresses, not hosts)

✓ 1. 2^4 ✓

2. $2^4 - 2$

3. 2^5

4. $2^5 - 2$

5. 2^6

6. $2^6 - 2$

3 How can I access another machine's localhost?

1. By adding an entry to `/etc/hosts`

2. By creating or modifying a file in `/etc/netplan/`

→ 3. It's not possible

✗ 4. By opening a listening socket in the target machine and accepting connections ✓

4 DNS is...

→ 1. A service that translates hostnames into IP addresses

✗ 2. A service that translates IP addresses into hostnames ✓

5 A docker container is...

1. Like a variable, but global

2. Like a script, but better

✓ 3. A kind of virtual machine

✓ 4. An isolated process ✓

6 What type of packages are used by Fedora?

1. apt
2. deb
3. yum

4. rpm

5. dnf

7 When I create a new user, the following file will be changed

1. /etc/users
2. /etc/passwd
3. /etc/shadow
4. /etc/light

8 A user on a Linux system can only be part of up to 2 groups: a group with the same name as the user and one additional group

1. true

2. false

9 When setting special permissions using the octal representation, the following value mappings are used:

1. SUID=4, SGID=2, sticky=1
2. SUID=4, sticky=2, SGID=1
3. SGID=4, SUID=2, sticky=1
4. SGID=4, sticky=2, SUID=1

10 How can I make sure the service X is started on boot?

1. systemctl start x.service
2. systemctl start --at-boot x.service
3. systemct enable x.service
4. systemct start --on-boot x.service

11 The network model in use today is called:

- 1. OSI Model
- ✓ 2. TCP/IP Model
- 3. UDP/IP Model

12 Is it possible for two different sockets to bind to the same IP address?

- ✗ 1. yes
- 2. no ✓

13 In the "3-2-1" rule for backups, the "2" part refers to...

- 1. 2 different mediums
- ✗ 2. 2 different locations
- 3. always take at least 2 copies
- 4. always store at least 2 copies on a different location

14 When building docker images, it's possible to use the cache from a previous build

- ✓ 1. true ✓
- 2. false

15 A docker image is:

- 1. another name for a container
- ✗ 2. the text file used to build a container, usually called Dockerfile ✓
- 3. connected to a tag, which can never change
- 4. the blueprint used to create containers

16 When starting a docker container from the cli (using docker run), I can create a bound volume using a relative path

- ✓ 1. true ✓
- 2. false ✓

17 When exposing ports from a docker container to the host, I have to use the same port on both the host and the container

✓ 1. true

2. false ✓

18 To achieve IPC (inter-process communication) on Linux, I can use signals, sockets and...

✓ 1. pipes ✓

2. redirects

3. processes

4. disk partitions

19 The GNU project is...

1. An ancestor of Linux

2. The original implementation of UNIX

3. A spin-off of Linux

4. A new, FOSS implementation of the UNIX system

5. None of the above

20 The following git commands need a remote to be set to properly work:

1. git open and git close

→ 2. git pull and git push

3. git pull, git push and git remote

✗ 4. git add, git commit, git push, git pull and git remote